Exercise Session 3: Bayesian Decision Theory, GMM, and Calibration

Automatic Image Analysis

Exercise Session 3

From probabilities to decisions: posterior inference, asymmetric clinical costs, Gaussian mixture models, and calibration.

Computer Vision and Remote Sensing Group, TU Berlin
Summer Semester 2026

- Bayesian decision theory from scratch.
- Density modelling with Gaussian mixtures.
- Reliability diagrams and Expected Calibration Error.

Roadmap

Decision theory

- Observations, classes, priors.
- Likelihoods and posteriors.
- Conditional risk.
- MAP as a special case.

Models and confidence

- Asymmetric clinical costs.
- Gaussian mixture models.
- BIC for model selection.
- Calibration and ECE.

Why this matters for the projects

Project connection

In the skin lesion triage project, the question is not only "which class is most likely?" but "which decision is acceptable under clinical risk?"

- False negatives can be much more costly than false positives.
- A classifier score is not automatically a calibrated probability.
- A transparent baseline should expose the prior, likelihood, cost matrix, and threshold.
- Deep models still need calibration checks before their confidence is trusted.

Bayesian setup

Objects

- Observation: x
- Classes: $\omega_1, \ldots, \omega_C$
- Prior: $P(\omega_i)$
- Likelihood: $p(x \mid \omega_i)$

Bayes rule

$$P(\omega_i \mid x) = \frac{p(x \mid \omega_i)P(\omega_i)}{\sum_{j=1}^{C} p(x \mid \omega_j)P(\omega_j)}$$

Interpretation

Posterior probability combines data evidence with prior class frequency.

From posterior to action

Conditional risk

For an action α_i, define

$$R(\alpha_i \mid x) = \sum_{j=1}^{C} \lambda(\alpha_i \mid \omega_j) P(\omega_j \mid x).$$

Bayes decision rule

Choose the action with the smallest conditional risk:

$$\alpha^*(x) = \arg\min_{\alpha_i} R(\alpha_i \mid x).$$

- The posterior says what is likely.
- The loss matrix says what mistakes cost.
- The decision rule combines both.

MAP as 0/1 loss

0/1 loss

$$\lambda(\alpha_i \mid \omega_j) = \begin{cases} 0, & i = j, \\ 1, & i \neq j. \end{cases}$$

Risk

$$R(\alpha_i \mid x) = 1 - P(\omega_i \mid x).$$

Result

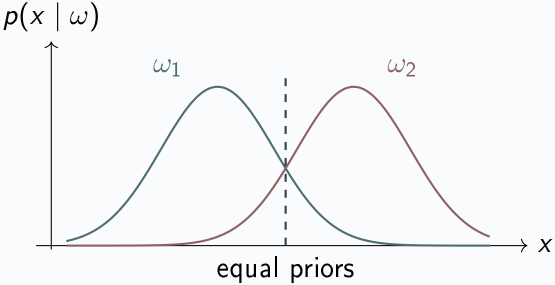
Minimizing risk is equivalent to maximizing posterior probability:

$$\alpha^*(x) = \arg\max_i P(\omega_i \mid x).$$

Name

This is the maximum a posteriori decision rule.

1D Gaussian example



Equal variance

With two Gaussians and equal variance, the boundary is where the posterior probabilities are equal.

Changing the prior

Increasing $P(\omega_2)$ shifts the boundary toward ω_1, because less likelihood evidence is needed to choose ω_2.

Asymmetric costs in skin lesion triage

Clinical motivation

Missing melanoma is more serious than sending a benign lesion for additional inspection.

Binary classes

- ω_M: melanoma
- ω_B: benign
- Action α_M: refer as melanoma
- Action α_B: treat as benign

Cost assumption

$$\lambda(\mathrm{FN}) = 5\,\lambda(\mathrm{FP})$$

False negatives are assigned five times the cost of false positives.

Decision threshold with asymmetric costs

Risks

If correct decisions have zero cost:

$$R(\alpha_M \mid x) = \lambda(\mathrm{FP})P(\omega_B \mid x), \qquad R(\alpha_B \mid x) = \lambda(\mathrm{FN})P(\omega_M \mid x).$$

Choose melanoma when

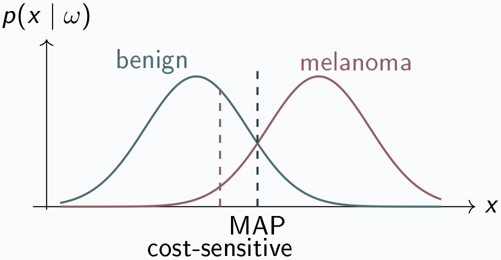
$$\lambda(\mathrm{FP})P(\omega_B \mid x) < \lambda(\mathrm{FN})P(\omega_M \mid x).$$

With $\lambda(\mathrm{FN}) = 5\lambda(\mathrm{FP})$:

$$P(\omega_M \mid x) > \frac{1}{6}.$$

- MAP with equal costs uses threshold 0.5.
- Asymmetric costs lower the melanoma threshold to reduce false negatives.

Boundary shift under asymmetric cost



Answer

It moves toward the benign class. More points are referred as melanoma, which reduces false negatives.

Practice

The cost matrix is chosen with domain experts. It is a policy choice, not something the classifier learns alone.

Gaussian mixture models

Flexible density estimator

A GMM models one class distribution as a weighted sum of Gaussian components:

$$p(x \mid \omega) = \sum_{m=1}^{K} w_m \mathcal{N}(x; \mu_m, \Sigma_m), \qquad \sum_{m=1}^{K} w_m = 1.$$

- One Gaussian is often too rigid for real image features.
- Several components can capture subgroups inside one class.
- For HAM10000, this can model colour/texture feature variation inside benign and melanoma examples.

EM intuition

E-step

Estimate soft component memberships:

$$r_{im} = \frac{w_m \mathcal{N}(x_i; \mu_m, \Sigma_m)}{\sum_{\ell=1}^{K} w_\ell \mathcal{N}(x_i; \mu_\ell, \Sigma_\ell)}.$$

M-step

Re-estimate parameters from responsibilities:

$$w_m = \frac{N_m}{n}, \qquad \mu_m = \frac{1}{N_m} \sum_i r_{im} x_i.$$

Loop

Repeat E-step and M-step until the likelihood stops improving.

Two EM iterations on five points

Toy data

Five 1D points: $x = \{1.0, 1.2, 1.4, 4.1, 4.4\}$, with $K = 2$ components.

Iteration 1

- Start with rough means near 1 and 4.
- E-step: left points get high responsibility for component 1.
- M-step: means move toward the weighted centres.

Iteration 2

- Responsibilities become sharper.
- Variances shrink around each cluster.
- Mixture weights approach 3/5 and 2/5.

Take-away

EM alternates between assigning points softly and updating the density model.

Choosing the number of mixture components

Problem

Larger K usually increases likelihood, but may overfit.

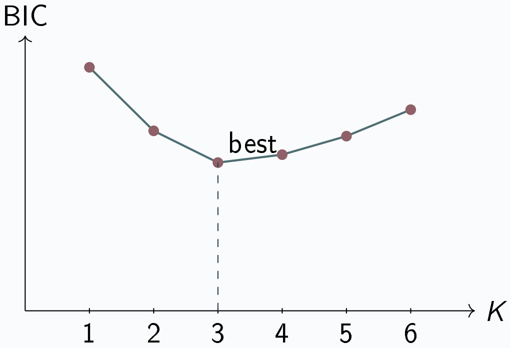
Bayesian Information Criterion

$$\mathrm{BIC} = k \ln n - 2 \ln \hat{L}$$

where k is the number of model parameters, n is the number of samples, and $\hat{L}$ is the maximized likelihood.

- Lower BIC is better.
- The penalty term discourages unnecessary components.
- Use the same feature space and split when comparing K.

Example BIC curve



Pre-run result

In the HAM10000 feature space, plot BIC against K and choose the lowest point.

Report

State the selected K, the feature representation, and whether the result is stable across random seeds.

Calibration: when confidence misleads

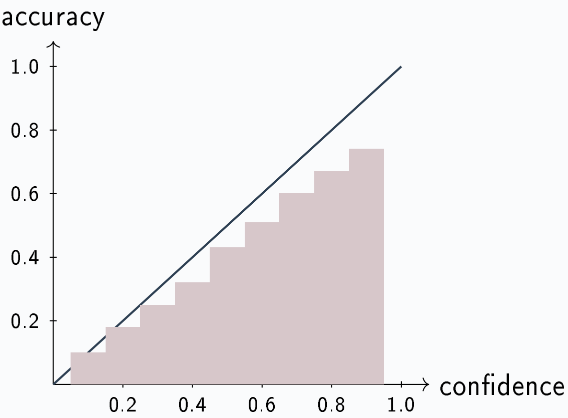
Definition

A binary classifier is calibrated if

$$P(\hat{Y} = 1 \mid \hat{p} = p) = p \quad \text{for all } p.$$

- If the model predicts 0.8 for 100 similar cases, about 80 should be positive.
- Accuracy and calibration are different properties.
- Deep networks can be accurate but overconfident.

Reliability diagram



Read the plot

The diagonal is perfect calibration. Bars below the diagonal indicate overconfidence.

Project use

Use about 15 bins for the reliability diagram and report what the model gets wrong about confidence.

Expected Calibration Error

Scalar summary

Split predictions into M confidence bins. Then

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left| \mathrm{acc}(B_m) - \mathrm{conf}(B_m) \right| .$$

- $\mathrm{acc}(B_m)$: empirical accuracy in bin m.
- $\mathrm{conf}(B_m)$: average predicted confidence in bin m.
- ECE is useful, but always show the reliability diagram too.

Temperature scaling preview

Logits

Neural networks often output logits z before the softmax.

$$\hat{p}_i = \frac{\exp(z_i)}{\sum_j \exp(z_j)}$$

Temperature scaling

$$\hat{p}_i(T) = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$

with $T > 1$ making predictions less extreme.

Important

Fit T on validation data. Do not use the test set to tune calibration.

Minimal implementation plan

1. Extract features from the skin lesion images or load cached features.
2. Split data into train, validation, and test sets with stratification.
3. Fit class-conditional Gaussian or GMM densities.
4. Compute posteriors using priors and likelihoods.
5. Compare MAP and cost-sensitive decisions.
6. Plot ROC curve, confusion matrix, reliability diagram, and ECE.

Keep it reproducible

Record random seed, split definition, feature type, number of mixture components, and cost matrix.

Optical flow: why it matters for Project E

Project connection

In the UCF-101 action recognition project, the classical baseline relies on dense optical flow to capture motion that frame-level appearance alone cannot encode.

- Some actions look similar as stills (e.g. standing vs. falling).
- Motion patterns discriminate many action classes.
- Dense flow provides a pixel-level motion descriptor before any learned representation is needed.
- Understanding the derivation helps explain where the method fails.

Brightness constancy and the flow constraint

Assumption

The intensity of a point moving in the image stays constant over a small time step:

$$I(x, y, t) = I(x + u\,\delta t,\ y + v\,\delta t,\ t + \delta t).$$

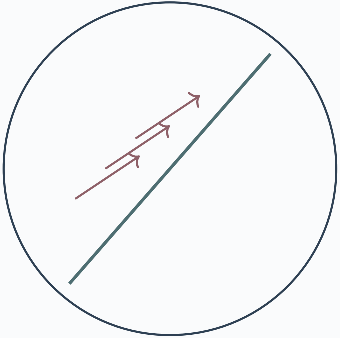
First-order Taylor expansion

$$I_x\,u + I_y\,v + I_t = 0,$$

where I_x, I_y are spatial gradients and I_t is the temporal gradient.

- One equation, two unknowns (u, v): the system is underdetermined.
- Additional constraints are required to obtain a unique solution.

The aperture problem



observed through aperture

Statement

Through a small aperture, only the component of motion perpendicular to an edge can be measured.

Consequence

The constraint equation allows infinitely many velocity vectors lying on the same line in (u, v) space.

Lucas–Kanade: resolving the ambiguity

Spatial coherence assumption

All pixels in a small patch $\mathcal{N}$ share the same flow (u, v). Stack one constraint per pixel:

$$\underbrace{\begin{pmatrix} I_x^{(1)} & I_y^{(1)} \\ \vdots & \vdots \\ I_x^{(n)} & I_y^{(n)} \end{pmatrix}}_{A} \begin{pmatrix} u \\ v \end{pmatrix} = -\underbrace{\begin{pmatrix} I_t^{(1)} \\ \vdots \\ I_t^{(n)} \end{pmatrix}}_{b}.$$

Least-squares solution

$$\begin{pmatrix} u \\ v \end{pmatrix} = (A^\top A)^{-1} A^\top b.$$

Well-conditioned only when $A^\top A$ has two large eigenvalues (texture in two directions).

When Lucas–Kanade is reliable

Reliable (corner-like regions)

- Both eigenvalues of $A^\top A$ large.
- Texture varies in x and y.

Unreliable

- One large eigenvalue: edge (aperture problem remains).
- Both small: homogeneous patch, no signal.

Practical limits

- Brightness constancy breaks under illumination changes.
- Large displacements require a coarse-to-fine pyramid.
- Dense variants (e.g. Farnebäck) extend LK to every pixel.

Dense optical flow as a feature

From pixels to descriptors

Dense flow produces a two-channel field (u, v) per frame pair. To obtain a fixed-size descriptor for classification:

1. Compute flow magnitude and angle at each pixel.
2. Build a histogram of flow orientations over the clip (analogous to HOG but temporal).
3. Concatenate histograms over spatial cells and time windows.
4. Train a multi-class SVM on the resulting descriptor.

Why SVM?

A linear or RBF kernel SVM on handcrafted flow features provides a transparent classical baseline before any deep model is applied.

Classical pipeline – Topic A: Leaf Species Recognition

1. Load Flavia images and segment to obtain binary leaf silhouettes.
2. Extract ordered boundary points from each silhouette.
3. Normalize the boundary: centre on centroid, fix scale, align start point.
4. Compute Fourier descriptors from the complex boundary signal; keep low-frequency coefficients.
5. Compute the seven Hu moments from the leaf region.
6. Concatenate Fourier coefficients and Hu moments into one feature vector per leaf.
7. Train naive Bayes and k-NN on a stratified train/test split.
8. Report per-class accuracy and confusion matrix for both classifiers.

Classical pipeline – Topic B: Skin Lesion Triage

1. Load HAM10000 images and convert to HSV colour space.
2. Compute per-channel histograms (H, S, V) and concatenate into one feature vector.
3. Split data into train, validation, and test sets with class stratification.
4. Fit a Gaussian mixture model per class (melanoma, benign) using EM.
5. Select the number of components K by minimizing BIC on the training set.
6. Compute class-conditional likelihoods and apply Bayes rule to get posteriors.
7. Derive decision threshold from asymmetric cost matrix; compare with MAP threshold.
8. Evaluate with ROC curve, confusion matrix, and reliability diagram.

Classical pipeline – Topic C: Matching Across Viewpoints

1. Load VGG benchmark image pairs and their ground-truth homographies.
2. Build a Gaussian scale-space pyramid for each image.
3. Compute the Difference-of-Gaussian (DoG) across consecutive scales.
4. Detect keypoints at DoG extrema; apply contrast and edge-response thresholds.
5. Assign a dominant gradient orientation to each keypoint.
6. Compute 128-dimensional SIFT descriptors at each keypoint location.
7. Match descriptors with nearest-neighbour ratio test (Lowe's ratio < 0.8).
8. Evaluate repeatability and matching score using the known homographies.

Classical pipeline – Topic D: STL-10 Linear Evaluation

1. Load the labelled portion of STL-10 (96×96 colour images, 10 classes).
2. Resize or crop to a fixed spatial resolution if needed.
3. Compute HOG features: choose cell size, block size, and number of orientation bins.
4. Flatten each image's HOG output into a single feature vector.
5. For each label fraction (e.g. 1%, 5%, 10%, 25%, 50%, 100% of training data):
   - Sample the fraction with class stratification.
   - Train a linear SVM on the sampled features.
   - Evaluate on the held-out test split.
6. Plot accuracy vs. label fraction as a learning curve.
7. Report the SVM hyperparameter C chosen by cross-validation at each fraction.

Classical pipeline – Topic E: Temporal Action Recognition

1. Load the 10-class UCF-101 subset; decode video clips to frames.
2. Sample consecutive frame pairs uniformly across each clip.
3. Compute dense optical flow for each pair (e.g. `cv2.calcOpticalFlowFarneback`).
4. Convert flow to polar form; compute magnitude and orientation per pixel.
5. Build a histogram of flow orientations over spatial cells and time windows.
6. Concatenate cell histograms into one fixed-size descriptor per clip.
7. Train a multi-class SVM (RBF or linear kernel); tune C by cross-validation.
8. Report accuracy, confusion matrix, and examples where motion is decisive.

Presentation 1 format

Format

Presentation 1 is short: **5 minutes** without Q and A. Use upto 5 slides to cover the essentials of your classical implementation plan.

May include

- Topic and research question.
- Dataset and subset.
- Classical baseline.
- Modern comparison plan.
- Evaluation metrics.

For this topic

- Cost matrix.
- Calibration plan.
- First risk or threshold experiment.
- Current blockers.